

CE201 INDUSTRIAL PRACTICE

TEAM PROJECT

INDIVIDUAL REFLECTIVE STATEMENT COVER SHEET

Module Code: **CE201**

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Student i.d. Number: **2316794**

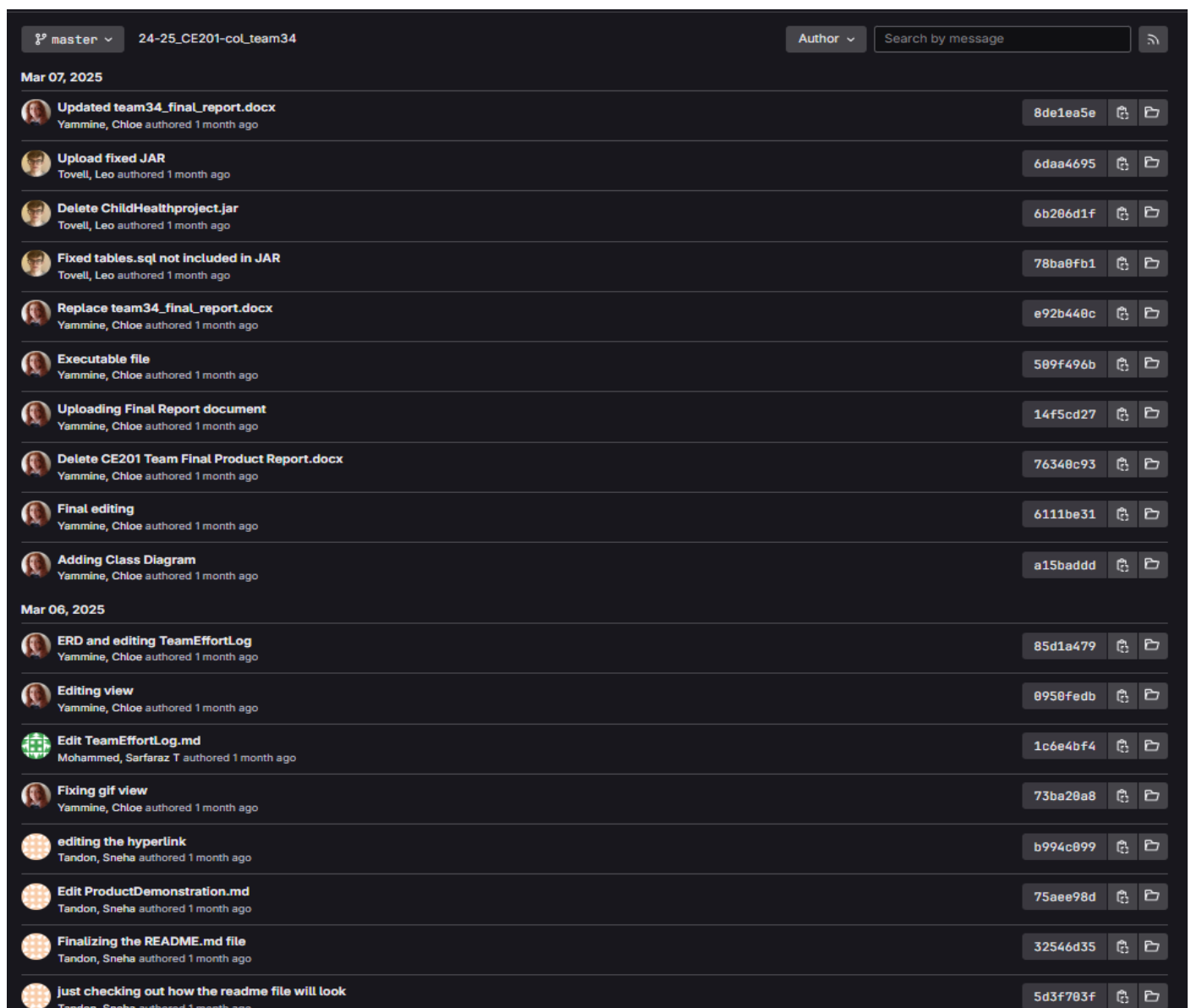
Team Number: **34**

Team Supervisor: **Dr Muhammad Tariq Sadiq**

Word Count: 718

The project, Child Health & Wellbeing Tracking, is an easy-going and data-driven solution designed to help stakeholders like parents, caretakers, or healthcare professionals to monitor essential aspects of a child's life. These features include sleep, feeding, diaper changes, medications, growth metrics and body temperature. To improve user experience, we allowed the main user to add multiple children's profiles and produce a PDF report with all input information. Each user's data is represented in graph format, showing trends over time.

In this project, I was the Database & Backend Integration Developer. I set up and optimized the SQL database, initially starting with MySQL and ensured compatibility with PostgreSQL. Working closely with Chloe and Leo, I established a strong database structure and integrated it seamlessly with Java. As a team, we ensured smooth database connectivity across devices. I also implemented PDF generation using iTextPDF and contributed to the basic structure of several application pages, laying groundwork for the GUI team. This project allowed me to enhance skills in SQL, Java, backend login, problem solving, team collaboration, version control (Git), and communication, ensuring both individual and team development.

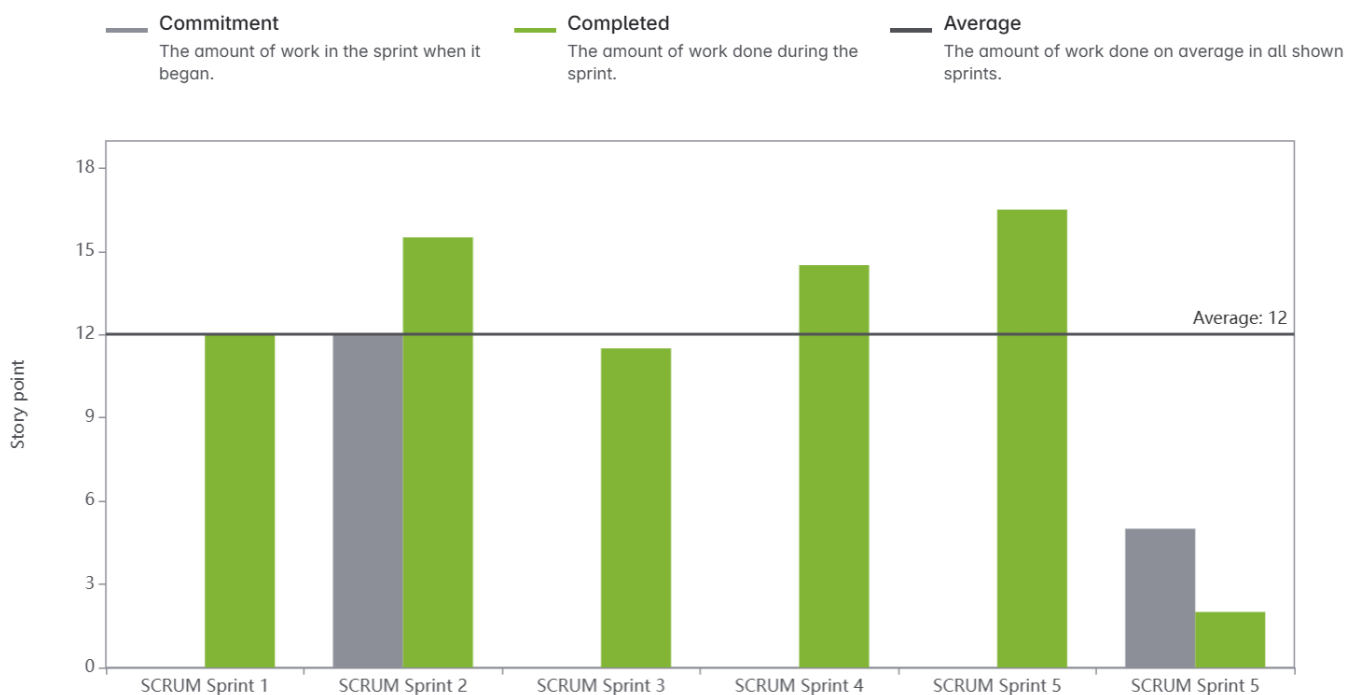


A strong collaborative spirit was displayed by our team. Leo focused on modularizing code, fixing bugs, and page structure. Sarfaraz translated user requirements into stories and conducted market research,

contributing to the market plan and report. Saif handled data validation, GUI development, and marketing. Chloe led in designing a normalized database and contributed to the front-end mock-ups and implementation. Denize developed graph functionalities, conducted unit testing with JUnit, and created wireframe designs that guided our product vision. The tools used helped us manage tasks efficiently, ensuring smooth sprints and timely integration. Presenting our project as a team helped us reflect strengths, weaknesses, and the importance of shared responsibility and communication.

Working on this project has significantly contributed to my professional development, empowering my technical and soft skills. It gave me hands-on experience in backend development, SQL optimization, and tools like PostgreSQL, iTextPDF, GitLab, and Jira. I developed stronger communication, time management, and problem-solving abilities, learning to adapt in a fast-paced, agile environment. The module enhanced my proficiency in Java and JUnit testing while strengthening my understanding of real-world software development cycles. It was a vital learning experience that helped me grow as a backend developer and team player.

The main tools used were Jira and GitLab. Jira helped us keep members accountable and meet deadlines, using sprint planning and burnout graphs. GitLab served as our version control, allowing structured collaboration through individual branches. Merge requests and issue tracking helped refine the project. These tools maintained efficient workflow, transparent communication, and supported our agile approach.

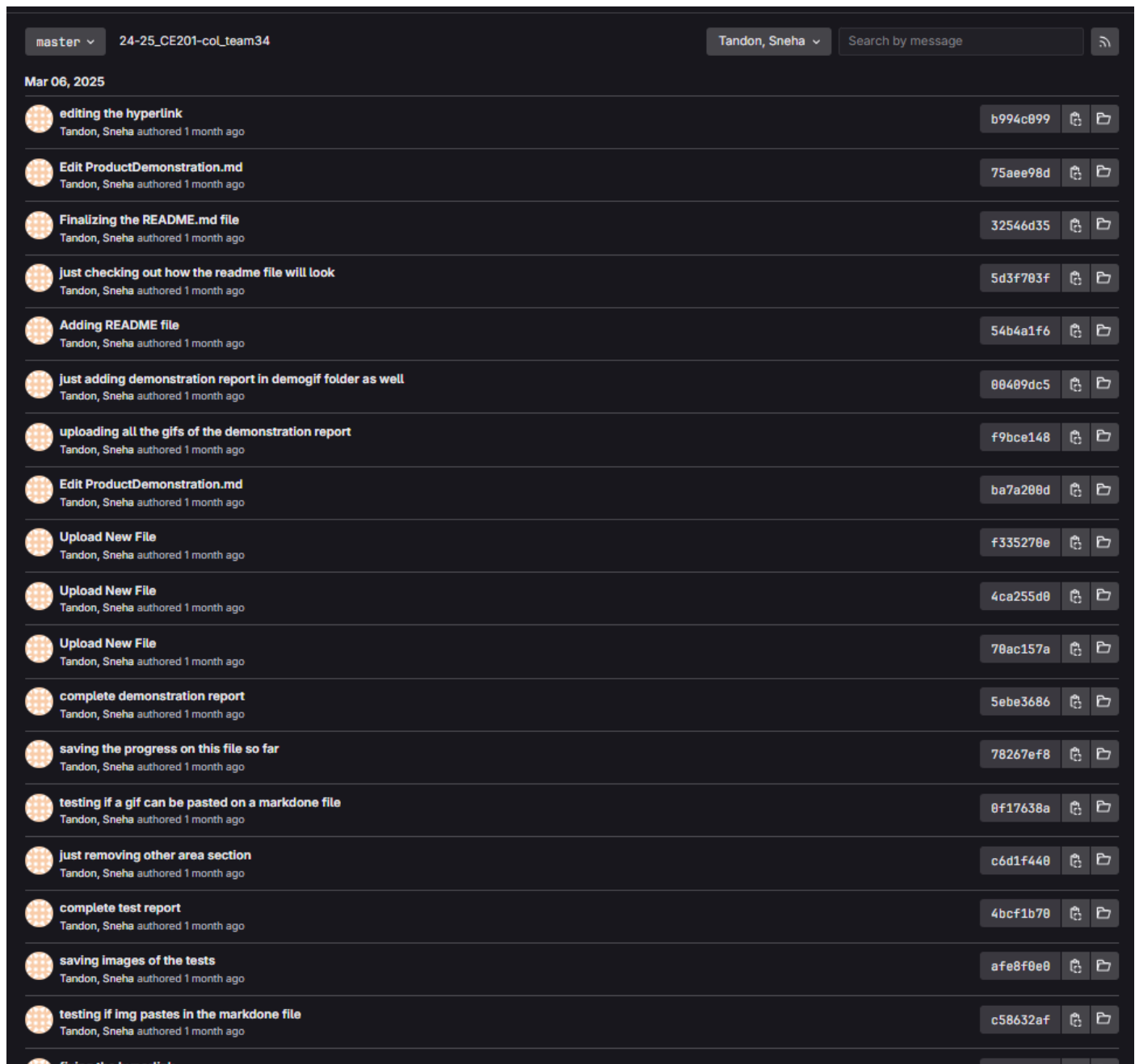


Working on this system inspired me to further pursue my personal creation. Building a real-world solution that impacts lives gave me confidence in my technical abilities. It made me realize how thoughtful, user-focused software can be influential. This journey sharpened my problem-solving and teamwork skills and sparked a better understanding of full stack development. It motivated me to continue my personal project – AI Interviewer, a smart platform that replicates real-world interview scenarios using AI.

Agile development played a crucial role in our project by promoting flexibility, improvement, and team collaboration. It helped ensure timely and budget-conscious completion. Using an iterative approach and regular feedback loops, we adapted to changing requirements and addressed issues. Agile improved our

code quality and ability to deliver functional components over this 6-month period. It helped us stay on track and prioritize user needs.

Through this project, I developed a strong understanding of project management using Jira and GitLab for planning, tracking, and collaboration. I applied system development techniques in backend programming, database integration, and PDF generation, and reflected on my contributions through effort logs. Working in a team improved my communication, teamwork and problem-solving. I also considered ethical and professional responsibilities when handling data. This experience enhanced my technical and research abilities, while helping me build a tailored CV and cover letter.



sneha	24-25_CE201-col_team34	Author	Create merge request	Search by message	
Mar 06, 2025					
	Merge remote-tracking branch 'origin/sneha' into sneha	Tandon, Sneha authored 1 month ago	258e54fb		
	testing pdf	Tandon, Sneha authored 1 month ago	23b959f6		
	Merge remote-tracking branch 'origin/sneha' into sneha	Tandon, Sneha authored 1 month ago	e22cd9f6		
	updating pdf page, making it extract data from database	Tandon, Sneha authored 1 month ago	3c11e8a6		
	updating pdf page	Tandon, Sneha authored 1 month ago	e73e982e		
	Merge remote-tracking branch 'origin/master' into sneha	Tandon, Sneha authored 1 month ago	9a08493f		
	Edit ProductImplementationReport.md	Tovell, Leo authored 1 month ago	36ebe6eb		
	Edit ProductImplementationReport.md	Tovell, Leo authored 1 month ago	280c7d36		
Mar 05, 2025					
	"TeamEffortLog updates 2 - Cayan Deniz Kucuktopana"	CayanDeniz authored 1 month ago	98c22f7e		
	"TeamEffortLog updates 2 - Cayan Deniz Kucuktopana"	CayanDeniz authored 1 month ago	5e77dfe9		
	"TeamEffortLog updates - Cayan Deniz Kucuktopana"	CayanDeniz authored 1 month ago	532d6e88		
	"ProductContext Ethical, Equality and Diversity Descriptions"	CayanDeniz authored 1 month ago	fda568bb		
	"ProductContext Legal Descriptions"	CayanDeniz authored 1 month ago	e21378de		
	Merge branch 'final-page-touches' into 'master'	Tovell, Leo authored 1 month ago	c76ecb2c		
	Finished adding the graph logic to each page (they now dynamically update)....	Tovell, Leo authored 1 month ago	8bcf388c		
	Final touches on ChangingNappyPage	Leo Tovell authored 1 month ago	84aacca9		
Mar 04, 2025					
	Merge branch 'chloe' into 'master'	Tovell, Leo authored 1 month ago	a96ac11f		

In summary, this project provided a significant learning opportunity, allowing me to implement technical skills in a practical environment while working collaboratively. Every stage played a role in my development as a programmer and teammate. It strengthened my backend, database, testing, and project management skills. Most importantly, it showed how impactful thoughtful software can be, inspiring me to continue building meaningful solutions.